

Final Sedentary-Time Analysis

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Purpose

This R Markdown file documents and reproduces the manuscript analysis for estimating waking sedentary time in NHANES 2011-2014 wrist accelerometer data. The project combines posture predictions from the CHAP wrist model with sleep/nonwear information from SWaN to estimate each participant's usual daily waking sedentary time.

The main goal is to produce survey-weighted, nationally representative estimates of waking sedentary time after excluding sleep/nonwear periods. The analysis calibrates a sleep/nonwear percentage cutoff against NHANES self-reported sleep duration, applies the selected cutoff at the primary analysis epoch, summarizes sedentary time to the participant level, and estimates overall and subgroup means using the NHANES survey design.

This file is also a reproducibility driver. It reads the analysis-ready accelerometer datasets from the `wristsed` data package, derives the analytic sample, recalibrates the sleep cutoff for the 1-hour SWaN primary analysis, generates manuscript tables and figures, and writes final outputs to `outputs/final_analysis/` and `figures/final_analysis/`.

Table 1: Primary SWaN 1-hour survey-weighted estimates.

Measure	Mean (h/day)	SE	95% CI lower	95% CI upper
Waking sedentary time	11.35	0.04	11.26	11.44
Waking time	17.09	0.05	16.99	17.19
Sleep/nonwear time	6.91	0.05	6.81	7.01
Self-reported sleep time	6.95	0.02	6.92	6.99

Packages

Analysis Configuration

The primary render uses the 1-hour SWaN epoch dataset. Sleep/nonwear removal is applied at the epoch level before participant-day sedentary totals are calculated. For figures that display hourly patterns, records are aggregated to hour only after the sleep/nonwear rule has been applied.

The primary analytic unit is the NHANES participant. Epoch-level accelerometer records are used to construct participant-day and participant-level sedentary time summaries, but survey-weighted estimation is performed at the participant level to avoid giving participants with more observed records or more valid days disproportionate influence.

Data Access

Reusable Functions

Self-Reported Sleep Benchmark

NHANES self-reported sleep duration is used as an external benchmark for selecting the accelerometer sleep/nonwear cutoff. The benchmark is estimated with the same combined 2011-2014 survey design used for the primary participant analysis.

Primary 1-Hour Cutoff And Participant Data

The final sleep/nonwear cutoff is selected from a fine-resolution objective curve. For each candidate cutoff, the analysis compares survey-weighted accelerometer-derived sleep/nonwear duration with the survey-weighted NHANES self-reported sleep benchmark. The selected cutoff is the value that minimizes the absolute difference between those two sleep-duration estimates. This fine-resolution selected cutoff, not the coarse sensitivity-grid cutoff, is used for the primary waking sedentary-time analysis.

The fine-resolution objective curve selected a 1-hour SWaN sleep/nonwear cutoff of 90.84%. At this cutoff, the survey-weighted accelerometer-derived sleep/nonwear estimate was 6.91 hours/day, and the absolute difference from the survey-weighted self-reported sleep benchmark was 0.043 hours/day.

Main Estimates

Survey-weighted estimates are computed at the participant level. Each participant contributes one usual daily waking sedentary-time estimate, one waking-time estimate, and one sleep/nonwear-time estimate to the survey design.

Note. Estimates are survey-weighted and expressed in hours per day. SE = standard error; CI = confidence interval. Pooled estimates combine the 2011-2012 and 2013-2014 NHANES cycles using recalibrated 4-year examination weights.

Table 2: Main survey-weighted estimates overall and by NHANES cycle.

Analysis group	Measure	Mean (h/day)	SE	95% CI
Pooled	Mean sedentary time	11.35	0.04	11.26-11.44
Pooled	Waking time	17.09	0.05	16.99-17.19
Pooled	Sleep/non-wear time	6.91	0.05	6.81-7.01
2011-2012 cycle	Mean sedentary time	11.46	0.06	11.34-11.58
2011-2012 cycle	Waking time	17.27	0.07	17.14-17.40
2011-2012 cycle	Sleep/non-wear time	6.73	0.07	6.60-6.86
2013-2014 cycle	Mean sedentary time	11.25	0.07	11.12-11.37
2013-2014 cycle	Waking time	16.92	0.08	16.76-17.08
2013-2014 cycle	Sleep/non-wear time	7.08	0.08	6.92-7.24

Justification for Combining Survey Cycles

To assess whether the two NHANES cycles could be pooled, we compared survey-weighted estimates of mean sedentary time between 2011-2012 and 2013-2014. The mean sedentary time was 11.46 hours/day (SE = 0.06) in 2011-2012 and 11.25 hours/day (SE = 0.07) in 2013-2014. The survey-weighted regression estimate of the difference was -0.21 hours/day ($p = 0.0251$). This difference is small relative to the overall mean, so the two cycles were combined for subsequent analyses using recalibrated 4-year examination weights.

NHANES Analytic Approach

The primary estimand in this manuscript is the population mean of each participant’s usual daily waking sedentary time. The analysis therefore treats the NHANES sample person, not the hour, as the survey-weighted analytic unit. This matches NHANES guidance: sample weights are assigned to sample persons and account for unequal probability of selection, nonresponse, and post-stratification to U.S. population control totals.

The repeated accelerometer records were still used fully to derive the participant-level exposure. For each candidate sleep cutoff, SWaN sleep/nonwear percentages determined whether an epoch was treated as waking time, CHAP sitting probabilities contributed fractional sedentary hours using the epoch duration, daily totals were computed, and valid days were averaged within participant. The person-level mean is therefore a summary of all available epoch- and daily-level information for that participant.

What is intentionally removed before the main survey estimation is the repeated record structure itself, because leaving epochs or days as separate weighted records would give participants with more observed records or valid days more influence than participants with fewer records. That would change the estimand from a population mean across people to a record-weighted mean across observed person-epochs or person-days.

Table 1

Notes: A full day is a 24-hour day represented by 24 hourly records; partial initiation and return days are excluded.

Epoch Sensitivity Table

Notes: Direct states do not apply a sleep/nonwear cutoff. For SWaN, direct states use the underlying SWaN wake, sleep, and non-wear classifications combined with CHAP waking-sedentary time. For NHANES, direct states use the official NHANES 1-minute sleep-state classifications combined with the same CHAP posture summaries. Epoch rows first aggregate sleep/nonwear states to the listed epoch length, then classify each whole epoch as waking when its sleep/nonwear percentage is below the calibrated cutoff; epochs at or above the cutoff are counted as sleep/non-wear.

Table 3: Analytic sample and wear-day distribution by NHANES cycle (2011-2014).

Full days	2011-2012	2013-2014	Total (2011-14)
1	29 (0.4)	32 (0.4)	61 (0.4)
2	37 (0.5)	31 (0.4)	68 (0.5)
3	48 (0.7)	30 (0.4)	78 (0.5)
4	37 (0.5)	30 (0.4)	67 (0.5)
5	56 (0.8)	30 (0.4)	86 (0.6)
6	68 (1.0)	43 (0.6)	111 (0.8)
7	6,606 (96.0)	7,531 (97.5)	14,137 (96.8)
Sample Analyzed (1-7)	6,881 (100.0)	7,727 (100.0)	14,608 (100.0)

Table 4: Epoch-level sensitivity analysis by sleep/wake detection algorithm.

Method	Row	Cutoff	Sedentary	Waking	Sleep/nonwear
SWaN	Direct states	-	6.52 (0.04)	12.09 (0.06)	11.91 (0.06)
SWaN	10m epoch	90.01	10.91 (0.05)	16.66 (0.06)	7.34 (0.06)
SWaN	30m epoch	91.67	11.39 (0.05)	17.13 (0.06)	6.87 (0.06)
SWaN	1h epoch	90.84	11.35 (0.04)	17.09 (0.05)	6.91 (0.05)
NHANES	Direct states	-	7.97 (0.04)	13.66 (0.06)	9.48 (0.06)
NHANES	10m epoch	90.01	11.05 (0.05)	16.79 (0.06)	7.21 (0.06)
NHANES	30m epoch	86.67	11.11 (0.05)	16.85 (0.06)	7.15 (0.06)
NHANES	1h epoch	85.01	11.21 (0.05)	16.94 (0.06)	7.06 (0.06)

Cutoff Sensitivity Grid

This step does not select the final sleep/nonwear cutoff. It evaluates a coarse grid of candidate cutoffs in 5-percentage-point increments to summarize how survey-weighted sleep/nonwear duration and waking sedentary time change across plausible decision rules. The final cutoff used in the primary analysis is the fine-resolution value selected above.

Demographic Missingness

Demographic, anthropometric, and survey-design variables are taken from the participant-level analysis datasets and reduced to one record per participant before survey estimation. Age is grouped as <18, 18-34, 35-44, 45-64, and ≥65 years. Race/ethnicity is retained using the NHANES-coded categories available in the analysis data. BMI is grouped as under/normal weight (BMI < 25), overweight (25 ≤ BMI < 30), and obesity (BMI ≥ 30). Participants with missing subgroup information are not recoded into another category; missing values are excluded only from the relevant subgroup/domain estimate.

These variables describe population heterogeneity in waking sedentary time, not repeated epoch-level covariates. Subgroup estimates for age, gender, race/ethnicity, and BMI are calculated from the participant-level survey design object using domain analyses. This preserves the NHANES design information for standard error estimation while estimating the mean for the population domain of interest.

The final analytic sample included 14,608 participants after applying accelerometer-data availability and survey-design requirements. Missingness in demographic and survey-design variables is shown relative to this analytic denominator so subgroup tables can be interpreted with the correct available sample for each domain.

Table 5: Missingness of demographic and survey design variables relative to the final analytic sample.

Variable	Analytic sample N	Missing n	Missing %	Available n
Age group	14608	0	0.00	14608
BMI category	14608	135	0.92	14473
Education	14608	0	0.00	14608
Gender	14608	0	0.00	14608
Marital status	14608	0	0.00	14608
Masked PSU	14608	0	0.00	14608
Masked stratum	14608	0	0.00	14608
Race/ethnicity	14608	0	0.00	14608
Recalibrated examination weight	14608	0	0.00	14608

Overview of Sedentary-Time Estimation

The analysis proceeded in the following steps:

1. **Start with wrist accelerometer data.** We used NHANES 2011-2012 and 2013-2014 wrist accelerometer files.
2. **Prepare the files for prediction models.** The raw files were organized by participant and day and converted into the formats needed by the posture and sleep/wear models upstream of this report.
3. **Identify sitting in 10-second epochs.** The CHAP/DeepPostures model classified each 10-second epoch as sitting or not sitting.
4. **Identify sleep and nonwear.** SWaN classified accelerometer records as wake wear, sleep, or nonwear; these predictions were summarized to the analysis epochs used in this report.
5. **Summarize predictions for analysis.** Epoch-level predictions were converted to epoch-level percentages for the primary analysis.
6. **Choose the sleep/nonwear cutoff.** Candidate cutoffs were evaluated by comparing survey-weighted accelerometer-derived sleep/nonwear duration with NHANES self-reported sleep duration.
7. **Keep only waking epochs for sedentary-time estimation.** Epochs below the selected sleep/nonwear cutoff were treated as waking epochs. Epochs at or above the cutoff were removed from waking sedentary-time estimation.
8. **Calculate sedentary hours for each day.** For each participant-day, sedentary percentages were multiplied by epoch duration and added across waking epochs.
9. **Average days within each participant.** Daily waking sedentary-time values were averaged across each participant’s valid wear days.
10. **Use the NHANES survey design.** Each participant’s usual daily waking sedentary time was analyzed using NHANES examination weights, masked variance strata, and masked primary sampling units.
11. **Estimate national and subgroup values.** Survey-weighted means and confidence intervals were estimated for the full population and for subgroups defined by age, gender, race/ethnicity, and BMI category.

Figure 1

Workflow for deriving hourly sedentary and sleep/nonwear summaries. (A) Raw 80 Hz accelerometer signal. (B) CHAP sedentary predictions and SWaN sleep/wake/nonwear classifications. (C) Hourly sedentary and sleep/nonwear percentages used to define waking sedentary time under candidate sleep cutoffs.

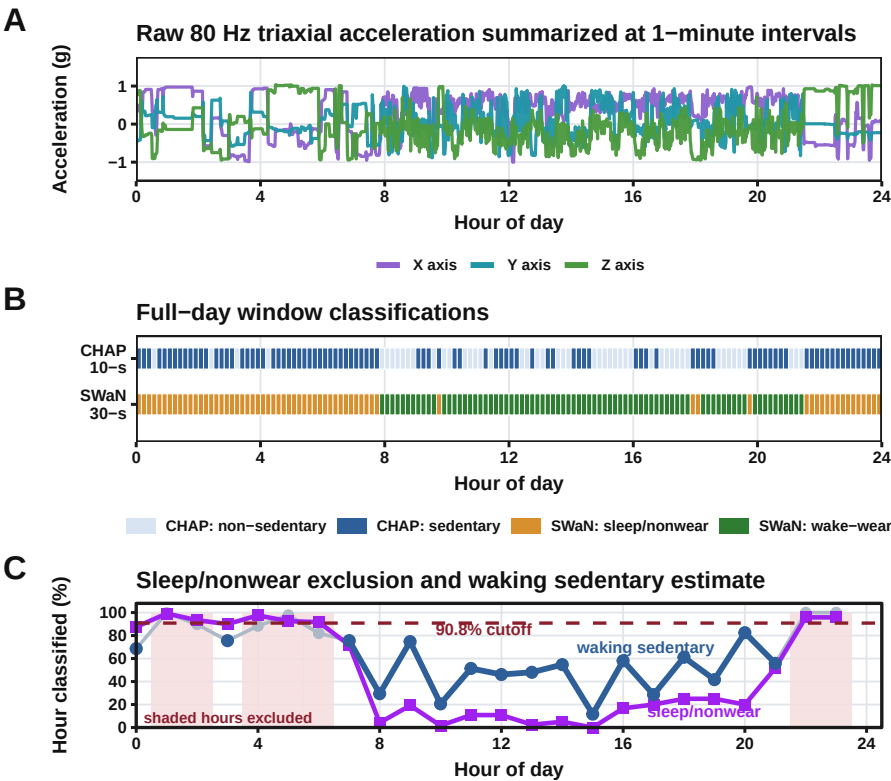


Table 2

This table places the current wrist-derived estimate beside prior NHANES sedentary-time estimates that used different devices, time periods, populations, or self-report instruments.

Figure 2

Survey-weighted hourly patterns of wrist-derived waking sedentary behavior are shown by day type after excluding records with sleep/nonwear greater than or equal to the selected 90.84% cutoff. Lines represent hourly means and shaded bands indicate 95% confidence intervals.

Table 6: Comparison of sedentary time estimates across studies by measurement method, datasets, and population.

Paper	Dataset & Population	Method	Estimate & Notes
This study	NHANES 2011-2014, wrist-worn 80 Hz accelerometer data; 14,608 participants, aged 3+ years.	CHAP wrist model combined with SWaN.	Sedentary time: 11.3 hours/day; 66.4% of non-sleep time. Mean waking time was 17.1 hours/day after excluding sleep.
Matthews et al. (2008)	NHANES 2003-2004; hip-worn ActiGraph data; 6,329 participants, aged 6+ years.	Intensity threshold of <100 counts/min.	Sedentary time: 7.7 hours/day; 54.9% of monitored time. Mean monitor wear time was 13.9 hours/day.
Yang et al. (2019)	NHANES 2001-2016; 51,896 children and adolescents.	Self-reported total sitting time from questionnaire.	Sedentary time: 6.4 hours/day in adults; 8.2 hours/day in adolescents. Sitting time increased in both groups from 2001 onward.
Ussery et al. (2018)	NHANES 2015-2016; 5,923 U.S. adults aged 18+ years.	Self-reported sitting time on a typical day.	~6.4 hours/day. Excludes sleeping time; 25.7% of adults reported sitting for more than 8 hours per day.

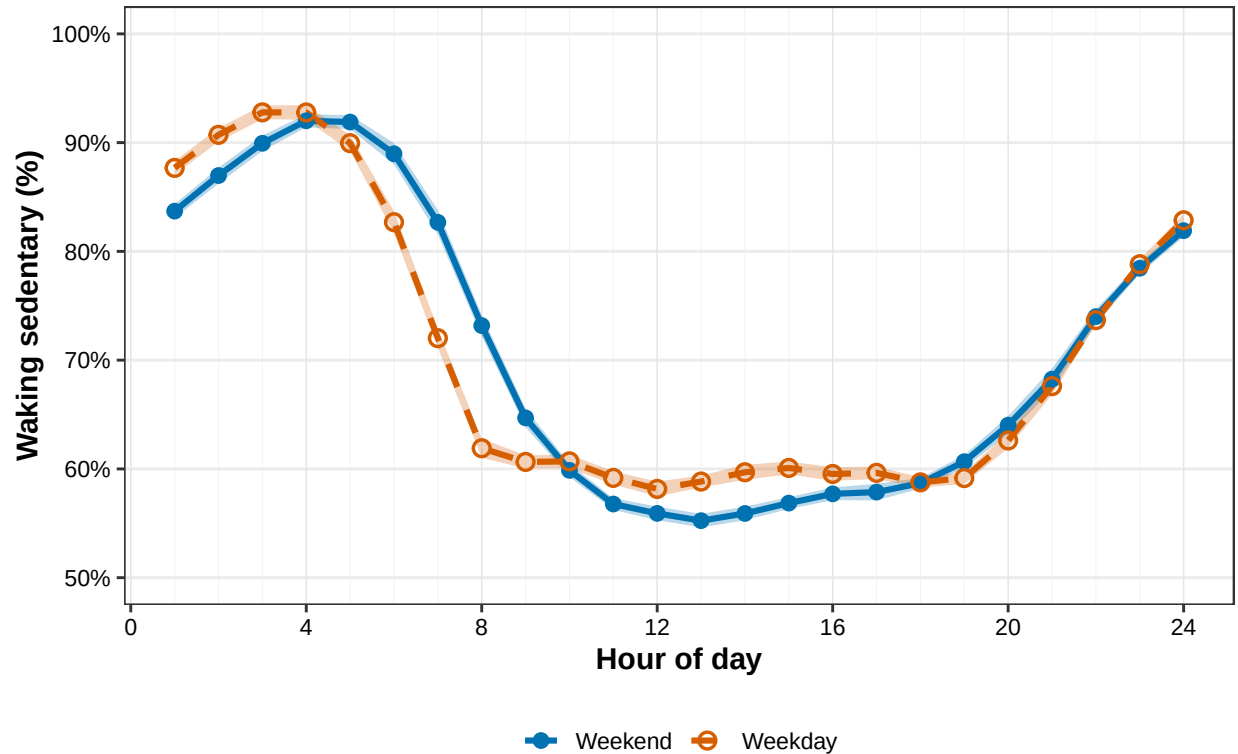


Table 7: Weighted sedentary behavior by gender and age, United States, 2011-2014.

Age group	Male n (%)	Male mean (95% CI)	Female n (%)	Female mean (95% CI)
All ages	7,142 (48.9%)	11.48 (11.35, 11.62)	7,466 (51.1%)	11.22 (11.12, 11.33)
<18	2,346 (16.1%)	10.54 (10.35, 10.73)	2,255 (15.4%)	10.20 (10.03, 10.36)
18-34	1,454 (10.0%)	10.88 (10.61, 11.15)	1,515 (10.4%)	10.60 (10.33, 10.87)
35-44	755 (5.2%)	11.08 (10.68, 11.48)	867 (5.9%)	10.79 (10.52, 11.06)
45-64	1,556 (10.7%)	11.93 (11.72, 12.14)	1,717 (11.8%)	11.69 (11.51, 11.86)
>=65	1,031 (7.1%)	13.52 (13.33, 13.72)	1,112 (7.6%)	12.89 (12.62, 13.15)

Table 8: Weighted mean sedentary time by race/ethnicity and BMI category, NHANES 2011-2014.

Race	Underweight/normal	Overweight	Obesity
All	10.78 (10.67, 10.90)	11.30 (11.15, 11.45)	12.08 (11.90, 12.26)
Mexican American	10.46 (10.22, 10.71)	10.35 (10.07, 10.64)	10.92 (10.68, 11.15)
NH Asian	10.99 (10.75, 11.24)	11.54 (11.20, 11.88)	11.80 (11.21, 12.40)
NH Black	11.06 (10.82, 11.31)	11.67 (11.28, 12.05)	12.43 (12.06, 12.80)
NH White	10.77 (10.61, 10.93)	11.43 (11.23, 11.63)	12.25 (12.01, 12.49)
Other Hispanic	10.46 (10.13, 10.78)	10.60 (10.22, 10.98)	11.13 (10.75, 11.50)
Other/Multi	11.02 (10.60, 11.44)	11.40 (10.32, 12.48)	13.03 (12.45, 13.61)

Figure 3

Distribution of participants by mean daily waking sedentary time and corresponding 24-hour behavioral composition. Panel A shows the proportion and unweighted number of participants in each sedentary-time category. Panel B shows the mean daily allocation to sedentary time, sleep/nonwear, and non-sedentary waking time within each category.

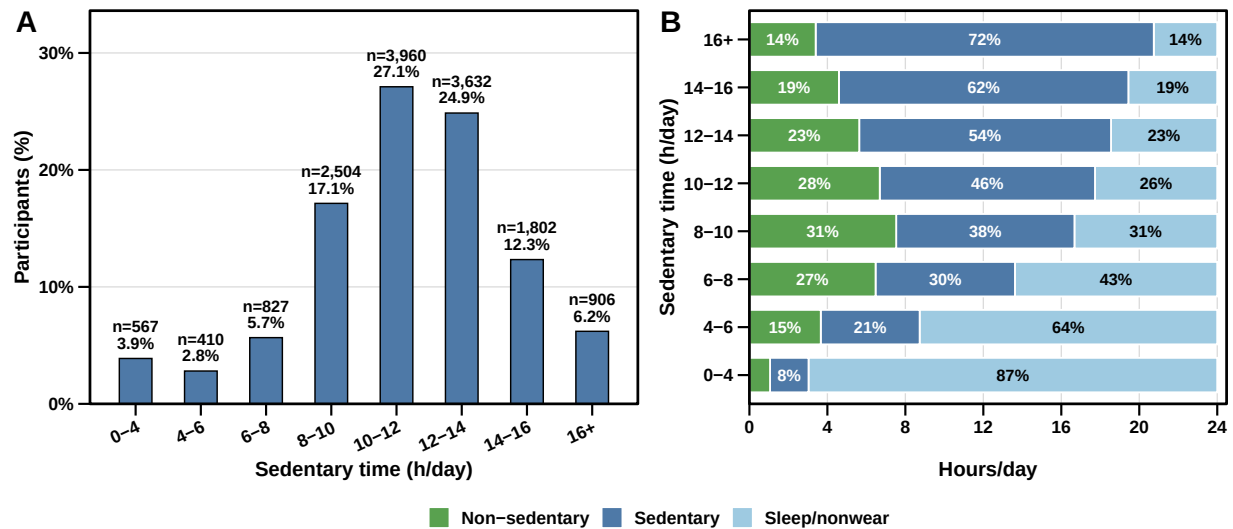


Table 9: Survey-weighted sedentary time by demographic subgroup, NHANES 2011-2014.

Subgroup	Category	N (%)	Mean sedentary time (95% CI), hours/day
Age group	<18	4,601 (31.5%)	10.37 (10.25, 10.49)
	18-34	2,969 (20.3%)	10.74 (10.51, 10.98)
	35-44	1,622 (11.1%)	10.93 (10.67, 11.19)
	45-64	3,273 (22.4%)	11.80 (11.68, 11.93)
	>=65	2,143 (14.7%)	13.17 (12.99, 13.35)
Gender	Male	7,142 (48.9%)	11.48 (11.35, 11.62)
	Female	7,466 (51.1%)	11.22 (11.12, 11.33)
Race/ethnicity	Mexican American	2,174 (14.9%)	10.61 (10.44, 10.79)
	NH Asian	1,673 (11.5%)	11.21 (11.01, 11.41)
	NH Black	3,628 (24.8%)	11.73 (11.49, 11.97)
	NH White	5,117 (35.0%)	11.45 (11.35, 11.55)
	Other Hispanic	1,431 (9.8%)	10.71 (10.46, 10.96)
	Other/Multi	585 (4.0%)	11.70 (11.24, 12.16)
BMI category	under/normal weight	6,972 (47.7%)	10.78 (10.67, 10.90)
	overweight	3,569 (24.4%)	11.30 (11.15, 11.45)
	obesity	3,932 (26.9%)	12.08 (11.90, 12.26)

Table 3

Table 4

Table 5

Sleep Distribution Diagnostic

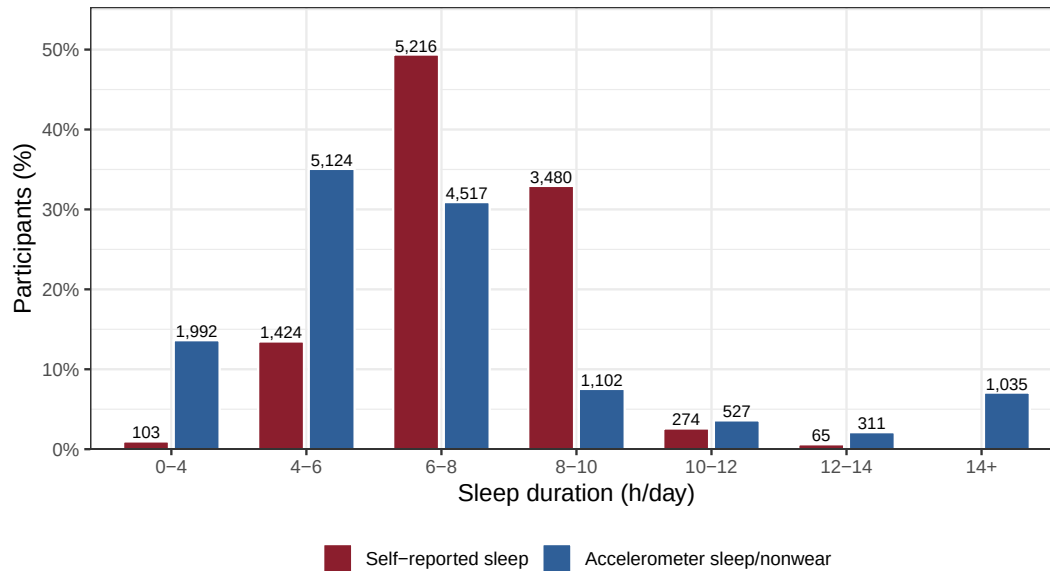
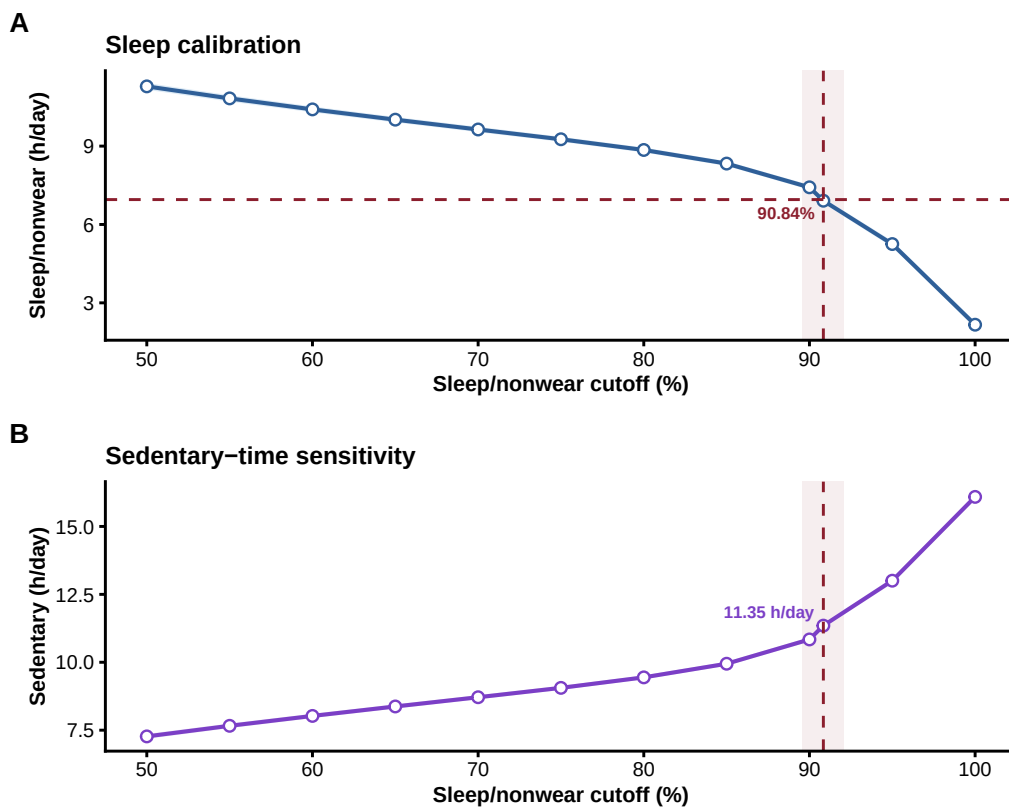


Figure 4

Sleep-cutoff calibration and sedentary-time sensitivity. Survey-weighted mean accelerometer-derived sleep/nonwear duration and waking sedentary time are shown across candidate sleep-percentage cutoffs. Shaded bands indicate 95% confidence intervals. The vertical dashed line marks the selected 90.84% cutoff,

calibrated to the self-reported sleep-duration benchmark.



Output Index

Table 10: Final analysis outputs.

Artifact	Path
Main results	outputs/final_analysis/results/main_results.csv
Main results by cycle	outputs/final_analysis/results/main_results_by_cycle.csv
Main estimates by cycle table	outputs/final_analysis/results/main_estimates_by_cycle_table.csv
Cycle comparison	outputs/final_analysis/results/cycle_comparison.csv
Table 1	outputs/final_analysis/tables/table1_wear_day_distribution.csv
Table 2	outputs/final_analysis/tables/table2_study_comparison.csv
Table 3	outputs/final_analysis/tables/table3_age_gender_sedentary.csv
Table 4	outputs/final_analysis/tables/table4_race_bmi_sedentary.csv
Table 5	outputs/final_analysis/tables/table5_marginal_subgroup_sedentary.csv
Epoch sensitivity table	outputs/final_analysis/tables/epoch_sensitivity_table.csv
Cutoff sensitivity grid	outputs/final_analysis/cutoff/cutoff_sensitivity_grid.csv
Objective curve	outputs/final_analysis/cutoff/objective_curve.csv
Selected cutoff	outputs/final_analysis/cutoff/selected_cutoff.csv
Demographic missingness	outputs/final_analysis/qc/demographic_missingness.csv
Analytic participant dataset	outputs/final_analysis/datasets/analytic_participant_sedentary.csv
Figure 1	figures/final_analysis/fig1_workflow.png
Figure 2	figures/final_analysis/figure2_hourly_waking_sedentary.png
Figure 3	figures/final_analysis/sedentary_distribution_with_24h_composition.png
Figure 4	figures/final_analysis/cutoff_sensi.png
Objective diagnostic	figures/final_analysis/objective_function_cutoff.png
Sleep distribution diagnostic	figures/final_analysis/sleep_distribution_comparison.png